

Algebra in Action: Adventures with Anna

July 2026

1. The ‘A4’ lectures and notes

On 3rd September 2026, I will give a couple of lectures on the topic of ‘Algebra in Action: Adventures with Anna’ (‘A4’). The abstract is:

ABSTRACT. Across scientific domains, researchers seek to understand complex systems by decomposing them into simpler parts. These lectures will discuss how to build a theoretical and practical framework for such problems, using the tools and insights of metric algebraic geometry. I will also report on enjoyable conversations and collaborations to study real-world systems. The goal is to initiate a conversation on open questions and future directions.

These (evolving) notes complement the lectures. Their goal, for now, is to help prepare the audience. There will be three main sections: the past, the present, and the future.

For even more background, the lecture notes for (the first half of) my graduate course ‘Algebraic fundamental of representing data’ are available at <https://seigal.github.io/module1.pdf> and <https://seigal.github.io/module2.pdf>. Another short related reference is [3].

Comments on all are welcome!

2. Introduction

Algebra in action means algebra for mathematical data science. That is, algebra that is devised or applied with the goal of answering a data analysis question. Examples of such questions may include those in biology, biomedicine, econometrics, literary studies, social sciences, etc. The questions have both trans-disciplinary and domain-specific aspects. We will reflect on what makes a data analysis question amenable, or not, to an algebraic point of view.

A central idea in the study of complex systems, such as those of the above applications, is the hypothesis of compositionality: observed variables (typically high-dimensional) can be described as combinations of (usually unobserved) simpler parts.

Formalized via random vectors, this hypothesis writes an observed random vector $\mathbf{x} \in \mathbb{R}^p$ as $\mathbf{x} = f(\mathbf{z})$ for a latent random vector $\mathbf{z} \in \mathbb{R}^q$, and a fixed unknown function $f : \mathbb{R}^q \rightarrow \mathbb{R}^p$. The special case where the combinations are linear replaces the map f by a fixed, unknown matrix $A \in \mathbb{R}^{p \times q}$. Recovering f or A and properties of $\mathbf{z}$ from samples of $\mathbf{x}$ is a factorization

problem. We will see that many ideas in data analysis can be thought of as such factorization problems. We begin with the earliest, most classical examples.

3. The past

3.1. What is a module? Multivariate data classically consists of a collection of vectors $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)} \in \mathbb{R}^p$. These each record measurements of p variables. Such data can be organized as a matrix $X \in \mathbb{R}^{n \times p}$: the n rows are the samples, the p columns index the variables, and entry X_{ij} is the value of variable j in sample i . For example:

Variable j	Sample i	X_{ij}
gene	cell	expression level of gene j in cell i
electrode	timepoint	voltage of electrode j at time i
pixel	image	intensity of pixel j in image i
word	document	count of word j in document i

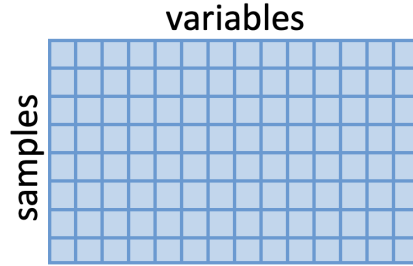


FIGURE 1. A data matrix, samples as rows, variables as columns

Data analysis seeks to uncover the building blocks that describe and explain such a matrix. Doing so amounts to fitting a model to the data. We view the vectors $\mathbf{x}^{(i)}, i = 1, \dots, n$ as samples of a vector of observed variables $\mathbf{x} = (x_1, \dots, x_p)^\top$. Writing observed variables as a linear combination of parts writes $\mathbf{x} = A\mathbf{z}$. We call $A \in \mathbb{R}^{p \times q}$ the mixing matrix: it encodes how the latent variables mix together to form the observed variables. At the level of samples, this says $\mathbf{x}^{(i)} = A\mathbf{z}^{(i)}$. Collecting together across all samples, this yields the matrix equation

$$X = ZA^\top,$$

where $X \in \mathbb{R}^{n \times p}$ is the data matrix and $Z \in \mathbb{R}^{n \times q}$ is the scores matrix. This product can also be written as a sum

$$X = \sum_{k=1}^q \mathbf{z}_k \mathbf{a}_k^\top,$$

where $\mathbf{z}_k$ and $\mathbf{a}_k$ are the k th columns of Z and A , respectively.

The fact that the matrix product is rank q records that the p observed variables have been expressed via q modules. By module, here, we don't mean in the commutative algebra sense, we mean a factor or process that describes the data. We don't get to see the latent variables themselves. Instead, we extract (in the rows of Z) how each module is expressed in a sample and (in the columns of A) how each variable participates in the module. In practice, one tends to think of this as being the module: you are where you show up, or you are what you consist of.

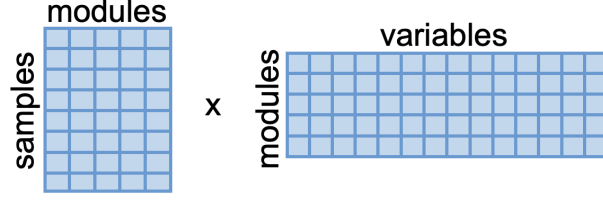


FIGURE 2. A low-rank decomposition $X = ZA^T$ of a data matrix writes it in terms of modules: the columns of Z record how much the module is expressed in each sample, while the rows of A^T record how much each observed variable participates in it. In this example, there are $n = 8$ samples, $p = 14$ variables, and $q = 5$ modules.

For example (going back to the table above): gene expression patterns can be encoded via gene modules, brain data via neural sources, images via features, and texts via their concepts and topics.

3.2. Principal component analysis (PCA). Assume that observed variables are described by uncorrelated latent variables $\mathbf{z}$ transformed via an orthogonal transformation, say V . Applying the covariance to the relationship $\mathbf{x} = V\mathbf{z}$ yields

$$\Sigma_{\mathbf{x}} = V\Sigma_{\mathbf{z}}V^T,$$

where the matrices $\Sigma_{\mathbf{x}}$ and $\Sigma_{\mathbf{z}}$ are the covariance matrices of $\mathbf{x}$ and $\mathbf{z}$, respectively. The covariance $\Sigma_{\mathbf{z}}$ is diagonal, since the entries of $\mathbf{z}$ are uncorrelated. Now we observe that this factorization is the eigendecomposition. Its solution exists and is generically unique by the spectral theorem.

The matrix $\Sigma_{\mathbf{x}}$ is the population covariance of the observed variables. In practice, we do not have access to it and instead use the sample covariance matrix:

DEFINITION 1 (Sample covariance). *Given data $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)} \in \mathbb{R}^p$ with sample mean $\bar{\mathbf{x}} = \frac{1}{n} \sum_i \mathbf{x}^{(i)}$, the sample covariance matrix is*

$$S = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}^{(i)} - \bar{\mathbf{x}})(\mathbf{x}^{(i)} - \bar{\mathbf{x}})^T.$$

PCA computes the eigendecomposition of the sample covariance matrix, truncated to some rank r . That is, it keeps the top r eigenvalues (and their corresponding eigenvectors) and sets the rest to 0. This is the best rank r approximation of the sample covariance matrix, by the Eckart-Young theorem.

Equivalently, PCA finds the singular value decomposition of the mean-centered data matrix X . That is, orthogonal matrices U and V and a diagonal matrix D such that $X = UDV^T$. The two formulations are equivalent, because

$$\frac{1}{n} X^T X = \frac{1}{n} (UDV^T)^T (UDV^T) = VD^2V^T.$$

3.3. Tensor decomposition. Above, our data consisted of points $\mathbf{x}^{(i)}$ in $\mathbb{R}^p$. Many datasets, though, carry multi-indexed structure, either on their space of variables, space of samples, or both. Examples include: fluorescence spectroscopy (samples $\times$ emission wavelengths $\times$ excitation wavelengths), color images (height $\times$ width $\times$ RGB channel), or neural

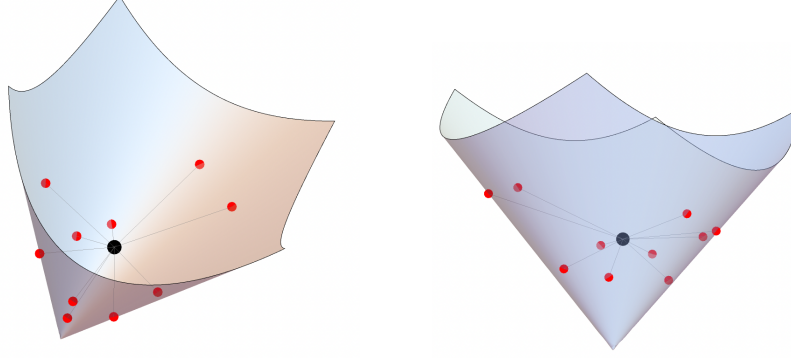


FIGURE 3. The 2×2 PSD cone (two views). Its boundary consists of matrices $\mathbf{v}\mathbf{v}^\top$ for $\mathbf{v} \in \mathbb{R}^2$. Here 10 mean-centered points (say $\mathbf{v}\mathbf{v}^\top$ for $\mathbf{v} \in \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$) on the boundary are shown in red. Their average (in black) is the sample covariance matrix of the data $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}$.

recordings (neuron $\times$ experiment $\times$ time). See [2]. As with a data matrix, one can decompose it into rank one pieces. Now each piece is a module. A module is its expression in each sample, the amount each variable participates in it, and its presence across each context.

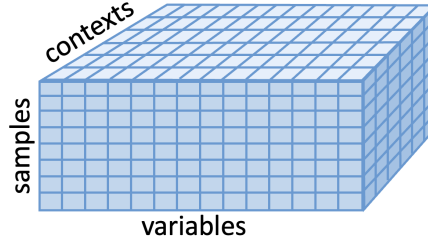


FIGURE 4. A data tensor, samples as rows, variables as columns, tubes as contexts

Theoretical and practical questions arise for optimal extraction of interpretable modules from tensors of data (i.e. for low rank approximation of tensors). For example: how should decompositions of different ranks be compared or combined? For PCA/SVD, the solution as truncation of the eigendecomposition to the top r eigenvalues mean the optimal solutions have a natural nested structure. This doesn't happen in general for tensors.

3.4. Independent component analysis (ICA). Independent Component Analysis (ICA) makes two changes relative to PCA in the equation $\mathbf{x} = A\mathbf{z}$: it strengthens the uncorrelated assumption on $\mathbf{z}$ to independence, and it removes the orthogonality restriction on A .

The classical motivating picture is the *cocktail party problem*: p microphones each record a mixture $x_i = \sum_j A_{ij}z_j$ of p underlying, independent sound sources z_j ; the goal is to recover A and the sources from the microphone recordings alone. This is an example of a blind source separation problem. One central way to recover the parameters in an ICA model is via tensor decomposition of cumulant tensors [1], as we now describe.

DEFINITION 2 (Moments). The d -th moment tensor of $\mathbf{z}$ is $\mathcal{M}_d(\mathbf{z}) := \mathbb{E}[\mathbf{z}^{\otimes d}] \in S^d(\mathbb{R}^q)$ with entries $\mathcal{M}_d(\mathbf{z})_{i_1 \dots i_d} = \mathbb{E}[z_{i_1} \cdots z_{i_d}]$.

The first moment is the mean: $\mathcal{M}_1(\mathbf{z}) = \mu$; the second moment is the matrix $\mathcal{M}_2(\mathbf{z}) = \mathbb{E}[\mathbf{z}\mathbf{z}^\top]$, which relates to the covariance by $\Sigma_{\mathbf{z}} = \mathcal{M}_2(\mathbf{z}) - \mu\mu^\top$.

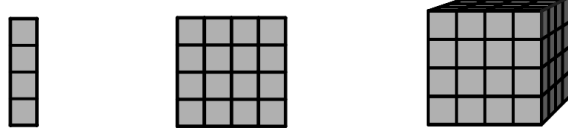


FIGURE 5. The first moment is a vector, the second is a matrix, the third is a third order tensor.

DEFINITION 3 (Moments and cumulants). For $\mathbf{t} = (t_1, \dots, t_q)^\top$, the moment generating function (mgf) of $\mathbf{z}$ is $M_{\mathbf{z}}(\mathbf{t}) = \mathbb{E}[\exp\langle \mathbf{t}, \mathbf{z} \rangle]$, and the cumulant generating function (cgf) is $K_{\mathbf{z}}(\mathbf{t}) = \log M_{\mathbf{z}}(\mathbf{t})$. We have

$$M_{\mathbf{z}}(\mathbf{t}) = \sum_{d \geq 0} \frac{1}{d!} \langle \mathbf{t}^{\otimes d}, \mathcal{M}_d(\mathbf{z}) \rangle.$$

and define the d -th cumulant tensor $\mathcal{K}_d(\mathbf{z}) \in S^d(\mathbb{R}^q)$ as the analogous coefficient of $K_{\mathbf{z}}$:

$$K_{\mathbf{z}}(\mathbf{t}) = \sum_{d \geq 1} \frac{1}{d!} \langle \mathbf{t}^{\otimes d}, \mathcal{K}_d(\mathbf{z}) \rangle.$$

DEFINITION 4 (The action of A on a symmetric tensor). For $A \in \mathbb{R}^{p \times q}$ and $S \in S^d(\mathbb{R}^q)$, define $A \bullet S \in S^d(\mathbb{R}^p)$ by

$$(A \bullet S)_{i_1 \dots i_d} = \sum_{\alpha_1, \dots, \alpha_d=1}^q S_{\alpha_1 \dots \alpha_d} A_{i_1 \alpha_1} \cdots A_{i_d \alpha_d}.$$

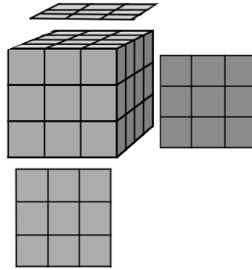


FIGURE 6. The action $A \bullet S$ multiplies a tensor by a matrix along each index.

For $d = 2$, $(A \bullet S)_{ij} = \sum_{\alpha\beta} S_{\alpha\beta} A_{i\alpha} A_{j\beta} = (ASA^\top)_{ij}$: this recovers the congruence action. The expression above is a product on the left and a sum on the right. For a matrix, we saw that a factorization can be viewed as a product or as a sum, too.

PROPOSITION 5 (Multilinearity). Let $\mathbf{x} = A\mathbf{z}$ for $A \in \mathbb{R}^{p \times q}$. Then $\mathcal{M}_d(\mathbf{x}) = A \bullet \mathcal{M}_d(\mathbf{z})$ and $\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z})$.

PROOF.

$$\mathcal{M}_d(\mathbf{x})_{i_1 \dots i_d} = \mathbb{E}[x_{i_1} \cdots x_{i_d}] = \mathbb{E}\left[\prod_{k=1}^d \sum_{\alpha_k} A_{i_k \alpha_k} z_{\alpha_k}\right] = \sum_{\alpha_1, \dots, \alpha_d} A_{i_1 \alpha_1} \cdots A_{i_d \alpha_d} \mathbb{E}[z_{\alpha_1} \cdots z_{\alpha_d}],$$

using linearity of expectation. The last factor is $\mathcal{M}_d(\mathbf{z})_{\alpha_1 \dots \alpha_d}$, giving $(A \bullet \mathcal{M}_d(\mathbf{z}))_{i_1 \dots i_d}$. The same proof applies to cgf Taylor coefficients $\square$

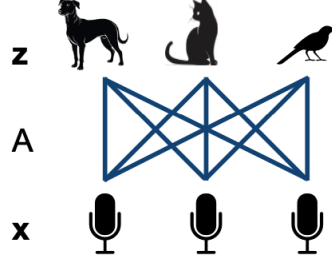


FIGURE 7. ICA for the cocktail party problem. Sources $\mathbf{z}$ are mixed together via mixing matrix A to produce observed variables $\mathbf{x}$.

PROPOSITION 6. Fix $d \geq 3$. Writing $A = [\mathbf{a}_1 \cdots \mathbf{a}_p]$ (columns) and $\lambda_i := \mathcal{K}_d(\mathbf{z})_{i,i,\dots,i}$ (the i -th source's order- d cumulant),

$$\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z}) = \sum_{i=1}^p \lambda_i \mathbf{a}_i^{\otimes d}.$$

PROOF. The cumulant tensors of mutually independent variables are diagonal. By the cumulant analog of multilinearity of moments (Proposition 5), $\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z}) = \sum_i \lambda_i (A \bullet \mathbf{e}_i^{\otimes d})$. We have $(A \bullet \mathbf{e}_i^{\otimes d})_{j_1 \dots j_d} = A_{j_1 i} \cdots A_{j_d i} = (\mathbf{a}_i^{\otimes d})_{j_1 \dots j_d}$, so $A \bullet \mathbf{e}_i^{\otimes d} = \mathbf{a}_i^{\otimes d}$. $\square$

4. The present

5. The future

Coming soon :-)

References

- [1] Pierre Comon and Christian Jutten. *Handbook of Blind Source Separation: Independent component analysis and applications*. Academic press, 2010.
- [2] Tamara G Kolda and Brett W Bader. Tensor decompositions and applications. *SIAM review*, 51(3):455–500, 2009.
- [3] Anna Seigal. Factorizations for data analysis. *SIAM News*, 58(07), 2025.